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EFFECT OF CATHINONE, THE ACTIVE CONSTITUENT OF KHAT ON CONDITIONED AVOIDANCE RESPONSE AND FEEDING BEHAVIOUR OF RATS

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ABSTRACT

Cathinone [(-)-α-aminopropiophenone is a psychoactive alkaloid of Khat (Cata edulis Forsk). The fresh leaves of this plant are customatrily chewed in Southern Arabia and Eastern Africa, to attain a state of euphoria. The present study was undertaken to study the effect of cathinone on conditioned avoidance responding (CAR) in rats. Studies on feeding behaviour after cathinone administration were also included with the aim of studying the relationship between the various behaviours induced and the dose of cathinone employed. The results of this study indicate that cathinone increases acquisition of CAR in naive male rats. Feeding behaviour studies indicated that cathinone suppresses food intake and causes reduction of body weights of rats without suppression of the desire to eat. The results of this study suggest the possibility of separation of the various effects of cathinone.

INTRODUCTION

Cathinone is a psychoactive alkaloid of khat (*Catha edulis* Forsk., Fam. Celesteraceae). The fresh leaves of this plant are customarily chewed in Southern Arabia and Eastern Africa, to attain a state of euphoria (Halbach, 1972). The chewer feels on increase of alterness and energy together with an enhanced depth of perception (Kalix, 1991). The actions of khat are mainly due to the alkaloid cathinone (Kalix, 1996). Khat is known to produce psychic dependence and has been classified as a drug of abuse by the World Health Organization (Nahas, 1981). Previous studies on khat have revealed its toxic nature (Kamel et al., 1967; De Hondt. et al., 1984; Islam et al., 1990). Other studies on behavioural (Foltin and Schuster 1981, 1982; Johanson and Schuster 1981) and neuro-chemical effects (Wagner et al., 1982) established (-)-cathinone, as a compound with sympathomimetic properties. Cathinone like other sympathomimetic amines is known as metabolic regulator (Knoll 1979; Nencine et al., 1984; Brenneisen and Geisshuster 1985), reduces food intake (Foltin and Schuster 1982) and decreases body weight (Zelger and Carlini, 1980).

Despite the extensive studies on the various behavioural aspects of cathinone, only few studies on avoidance behaviour have previously been reported. The present study was therefore, undertaken to study the effect of cathinone on conditioned avoidance responding in rats. In addition, the effects of cathinone on feeding behaviour were also studied. This is aimed at studying whether these animal behaviours and the dose of cathinone employed are related or not.

MATERIALS AND METHODS

Animals:

Naive white male Sprague-Dawley rats, (200-250 g) obtained from The Animal Care Centre, King Saud University, were used. The animals were housed in community cages. Food and water were available ad libitum in the home cage except when otherwise specified. The environment was controlled at a temperature range of 24+1°C with a 12 h and 14:00 h. Each animal was used only once.

Apparatus:

A computerized reflex conditioning system (Model: Reflex-16/Apple Columbus Instruments, Ohio, USA) was used to study the acquisition of conditioned avoidance response (CAR) in rats. The system consisted of 3 shuttle boxes made of acrylic plastic, and had a control unit. The control unit was used to send separate commands to activate a light stimulus, acoustical stimulus and/or deliver electric shock via stainless steel bars on each cage floor. Each shuttle box was divided into two compartments by a cage dividing door and was equipped with infra-red beam cells on each side of the dividing door. Crossing of the animal from one to the other compartment of the shuttle box would interrupt the beams and terminate the electric shock.

Feeding behaviour was studied using animal feeding monitor (Model FM04, Columbus Instruments, Ohio, USA). The feeding monitor was an optical-based system equipped with invisible infra-red beams and a steel animal feeder. Each time the animal interrupted the infra-red beam to eat, a single pulse was recorded. These pulses provided the number of times the animal visited the feeder for food. An internal clock in the control unit of the system measured the time that animal spent eating.

Procedure:

The CAR acquisition test consisted of one session of 100 trials for each rat. Animals were divided randomly into groups of 6. Each group was assigned to drug or saline treatments. Drugs were administered, i.p., 30 min before testing. Each animal (S) was placed in the shuttle box and allowed 45 min. of free exploration before the first trial was started. Each trial commenced with the buzzer and light (CS) which remained on for 10 seconds. Simultaneously with cessation of the CS, the shock (US) was delivered via the cage floor. The US was left on until the rat moved to the opposite compartment of the box or until 10 seconds elapsed. The intertrial interval (ITI) began with an avoidance or escape response, or at the cutoff time. If the S moved to the opposite compartment of the box during the CS interval, the response as scored as an avoidance. If the animal responded while the shock was being presented, the response as scored as an escape. Degree of acquisition for each group was expressed as mean percentage of avoidance responses in 10 blocks of 10 trials each.

In feeding behaviour studies the rats were deprived of food at 8.30 A.M. the previous day though they were allowed free access to water. On the experimental day, animals were placed individually in cages. Injections were made 30 min before testing. The number of feedings and the time spent feeding were recorded at intervals of 1 hr. for a period of 5 hours. Similarly treated groups of animals were used to measure the food intake at 1 hr. intervals for a period of 5 hours. Groups of 6 animals were used in each test.

Separate groups of rats were used in body weight experiments. The body weight of each rat was recorded on the first day and this was followed by an injection of the test drug (or saline). Injections were repeated on alternate days until 6 injections were given to each rat. Three groups of male rats were used. The first group was saline-treated and consisted of 12 animals. The other two groups were cathinone-treated (doses of 10 and 20 mg/kg respectively, were used), and consisted initially of 6 animals each. The body weight of each animal on each experimental day was recorded and was calculated as a percentage of the initial body weight registered before the start of each experimental day.

Cathinone was synthesized following the method of Al-Meshal et al. (1987). It was dissolved in 0.9% NaCl. Injection volume was 1 ml/kg. Doses were expressed as mg/kg. All injections were given intraperitionally.

Statistics:

One way ANOVA was used for the analysis of CAR acquisition data. Where significance was indicated, post-hoc comparisons were performed with Turkey-Kramer tests. Also, the Mann-Whitney U-test was used for the evaluation of the data of the other experiments; level of significance was taken at $P < 0.05$.

RESULTS

Effect of cathinone on CAR acquisition:

Fig. 1 shows the effects of various does of cathinone on the acquisition of CAR in rats. A one-way ANOVA for all data revealed a significant difference between the various treatments [$F(4.25 = 5.54$, $P < 0.05)$] on CAR acquisition. However, the lowest dose of cathinone (1 mg/kg) did not influence CAR acquisition. Cathinone (2 mg/kg) significantly enhanced acquisition of CAR. Although a dose of 4 mg/kg, apparently enhanced acquisition of CAR yet the effect was not significant. In contrast, the highest dose (10 mg/kg) caused complete inhibition of CAR acquisition. Comparing the mean response for the dose level of 2 mg/kg with its corresponding control at each trial block, the effect was significantly higher at the third and fifth to the eighth trial blocks ($P < 0.05$).

Effect of cathinone on feeding behaviour:

Fig. 2 depicts the number of feedings against time observed after various doses of cathinone in rats. All the doses of cathinone (5, 10 and 20 mg/kg) significantly decreased the number of feedings at 1 hr. However, only the dose of 20 mg/kg decreased the number of feedings at2 hours. A significant increase in the number of feedings was shown by cathinone (5 mg/kg) at 3 hours and by all doses of cathinone at 5 hours.

The total time spent by the animals trying to reach the food during the whole 5-hour observation is shown in Fig.3. All the doses of cathinone significantly increased the time spent feeding in comparison to saline-treated animals.

The effect of cathinone on food intake is shown in Fig.4. In contrast to the above experiments on feeding behaviour, here the fasted animals are allowed to reach the food. Cathinone (5 mg/kg) significantly decreased the food intake during the first hour of the 5-hour observation period. Also, the relatively higher dose of cathinone (i.e. 10 mg/kg) significantly decreased the food intake but for a longer than that caused by cathinone, 5 mg/kg.

Effect of chronic treatment with cathinone on body weight:

Fig. 5 shows the effects of 2 doses of cathinone given on alternate days (for a total of 6 injections) to male rats on their body weights. The smaller dose of 10 mg/kg of cathinone apparently reduced body weight in comparison to animals treated during the same period with saline. The reduction in body weight was however, significant only on day (5) of the experiment. The higher dose of cathinone (i.e. 20 mg/kg), on the other hand showed a clear-cut reduction on body weight of rats at days (5), (8), (10) and (12) of the experiment. Chronic treatment with cathinone at these doses seemed to be toxic. Out of the six animals treated with cathinone (20 mg/kg) 2 animals died on day (5) of the experiment. Also one animal died on day (12) and another died on day (15) from the group of animals treated with cathinone (10 mg/kg).

It has been observed that pretreatment with cathinone did not hinder the normal development of animals once cathinone injections were stopped. It appeared therefore, that the reducing effect of cathinone on body weight was reversible.

DISCUSSION

The results of the present study indicated that cathinone increased acquisition of CAR in naive male rats. The effect of cathinone on CAR acquisition was dose specific since it was only shown by cathinone (2 mg/kg). Higher doses of cathinone either reduced or suppressed (4 and 10 mg/kg, respectively) the acquisition of CAR. It appeared that a bell-shaped dose-response relationship is exhibited to cathinone in this behaviour. This bell-shaped relationship is also characteristic of amphetamine in a number of behavioural aspects (Taylor and Synder, 1970).

It has been indicated that the effects of cathinone are dopaminergically mediated (Huang and Wilson, 1984; Calcagnetti and Schecheter, 1992). Therefore, it seems likely that central dopaminergic systems are involved in CAR acquisition induced by cathinone.

Stimulation of locomotor activity by cathinone has been reported in rats (Yanagita, 1979) at a dose range close to that of amphetamine. Interestingly the increased acquisition of CAR produced by cathinone was shown by a dose which has no/or little effect on locomotor activity.

From feeding behaviour studies it was observed that the animals did not approach food for a brief period but they spent a large time after cathinone treatment trying to reach food. Like the amphetamines and other anorectic agents, cathinone depressed food intake as shown previously (Knoll, 1979; Zelger and Carlini, 1980). Interestingly, cathinone did not suppress the desire to eat. The body weight of animals was significantly reduced after prolonged cathinone administration. However, on stoppage of cathinone, the animals regained body weight quickly. Thus, the effect of cathinone on body weight, appeared to be reversible.

The enhancement of CAR acquisition by cathinone was seen at relatively lower doses than those influencing feeding behaviour. Therefore it is possible to separate the effect of cathinone on CAR acquisition from its unwanted deleterious effects such as increased locomotion, stereotypy, suppression of food intake and loss of body weight Khat chewers probably do not experience many of the harmful effects of cathinone because of these separable effects.

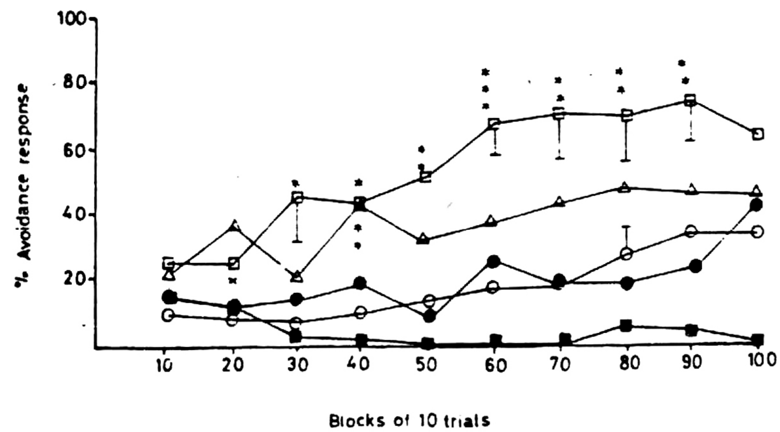

Fig. 1. **Effect of various doses of cathinone on the acquisition of conditioned avoidance in rats.**

O Saline (Control) ● 1 mg/kg □ 2 mg/kg

Δ 4 mg/kg ■ 10 mg/kg

***P < 0.05, ** P < 0.01, *** P < 0.001.**

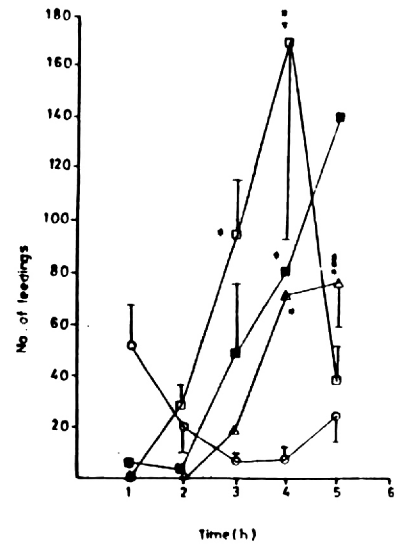

Fig. 2. Effect of various doses of cathinone on the number of feedings of fasted rats.

O Saline (Control) □ 5 mg/kg ■ 10 mg/kg
Δ 20 mg/kg
* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

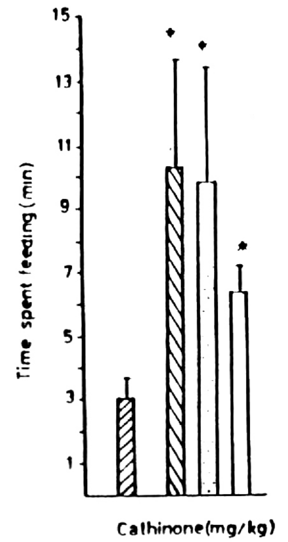

Fig. 3. Effect of various doses of cathinone on the time spent feeding of fasted rats.

Saline (Control) 5 mg/kg 10 mg/kg
□ 20 mg/kg
* $P < 0.05$.

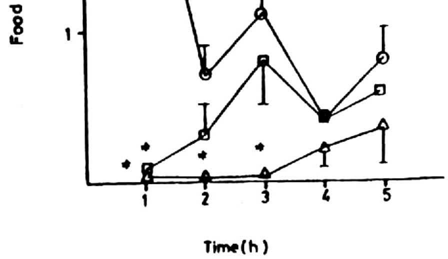

Fig. 4. The time course effect of various doses of cathinone on food intake of fasted rats.

O Saline (Control) □ 5 mg/kg) Δ 10 mg/kg
* $P < 0.05$.

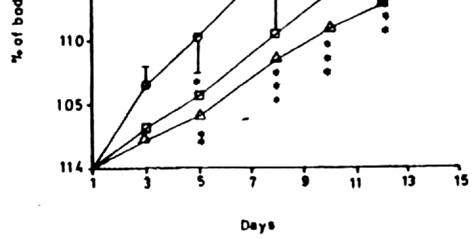

Fig. 5. The effect of chronic treatment with cathinone on body weight of rats.

O Saline (Control) □ 5 mg/kg Δ 20 mg/kg
* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

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